

DATABASE MANAGEMENT SYSTEMS (CS301)

Functional Dependencies

Learning Objectives

Understand functional dependencies, their types, Armstrong's axioms, attribute closure, and their role in database normalization.

Introduction to Functional Dependency

A Functional Dependency (FD) is a relationship between attributes in a relation where one attribute uniquely determines another attribute.

Definition

If attribute Y is determined by attribute X, then it is written as $X \rightarrow Y$ and read as 'X functionally determines Y'.

Example

Student(StudentID, Name, Branch)

StudentID $\rightarrow$ Name
StudentID $\rightarrow$ Branch

Since StudentID uniquely identifies a student, it determines other attributes.

Types of Functional Dependencies

Functional dependencies are classified based on how attributes depend on each other.

Full Functional Dependency

An attribute is fully dependent on a key when it depends on the entire key and not on any subset of the key.

Partial Dependency

A partial dependency exists when a non-key attribute depends on only part of a composite key.

Transitive Dependency

A transitive dependency occurs when a non-key attribute depends on another non-key attribute.

Trivial Dependency

A dependency $X \rightarrow Y$ is trivial if Y is a subset of X.

Non-Trivial Dependency

A dependency $X \rightarrow Y$ is non-trivial if Y is not a subset of X.

Armstrong's Axioms

Reflexivity: If $Y \subseteq X$, then $X \rightarrow Y$

Augmentation: If $X \rightarrow Y$, then $XZ \rightarrow YZ$

Transitivity: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$

Attribute Closure

Attribute closure is the set of all attributes that can be functionally determined from a given set of attributes.

Importance of Functional Dependencies

- Helps identify keys
- Supports normalization
- Reduces redundancy
- Improves database design

Applications

- Database Normalization
- Schema Design
- Data Integrity Enforcement
- Query Optimization

Exam Important Questions

1. Define Functional Dependency.
2. Explain Full and Partial Dependency.
3. What is Transitive Dependency?
4. State Armstrong's Axioms.
5. Explain Attribute Closure with example.

Summary

Functional Dependencies form the foundation of normalization and database design. Understanding different dependency types and Armstrong's axioms is essential for building efficient relational databases.